

DeepThyro: Evaluating NMS-Free YOLO26 Architectures for High-Resolution Thyroid Nodule Detection in Ultrasound

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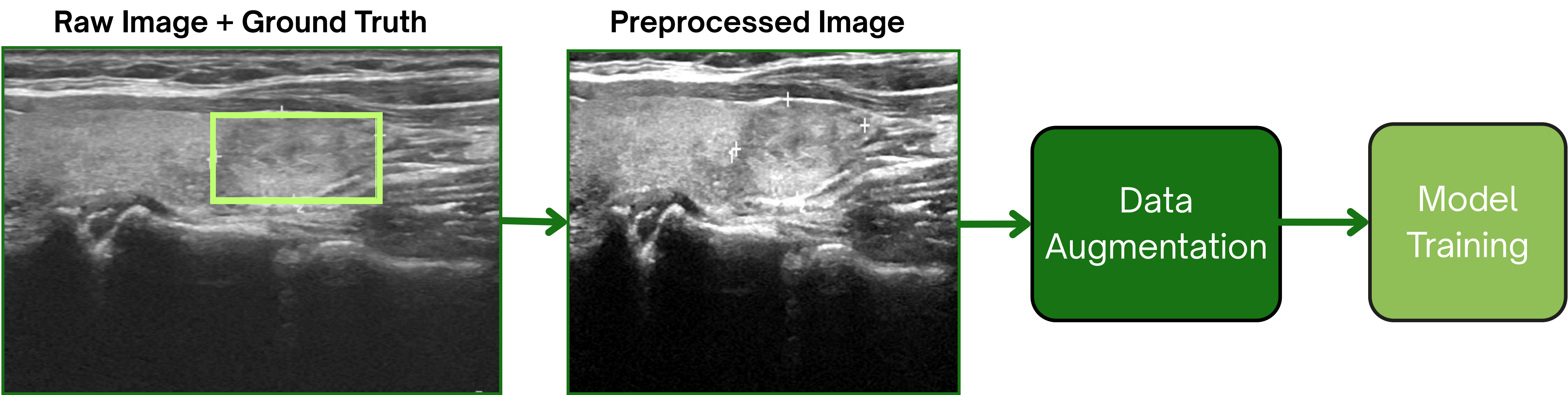
Problem

- **The Challenge:** Subtle micro-calcifications and irregular margins are difficult to detect in low-contrast ultrasounds.
- **The Limitation:** Standard image downsampling (e.g., 640x640 px) destroys these critical fine-grained details, drastically reducing diagnostic reliability.
- **The Bottleneck:** This loss of detail, combined with the extreme size difference between micro-nodules and large tissues, causes severe training instability and convergence failure at standard resolutions.

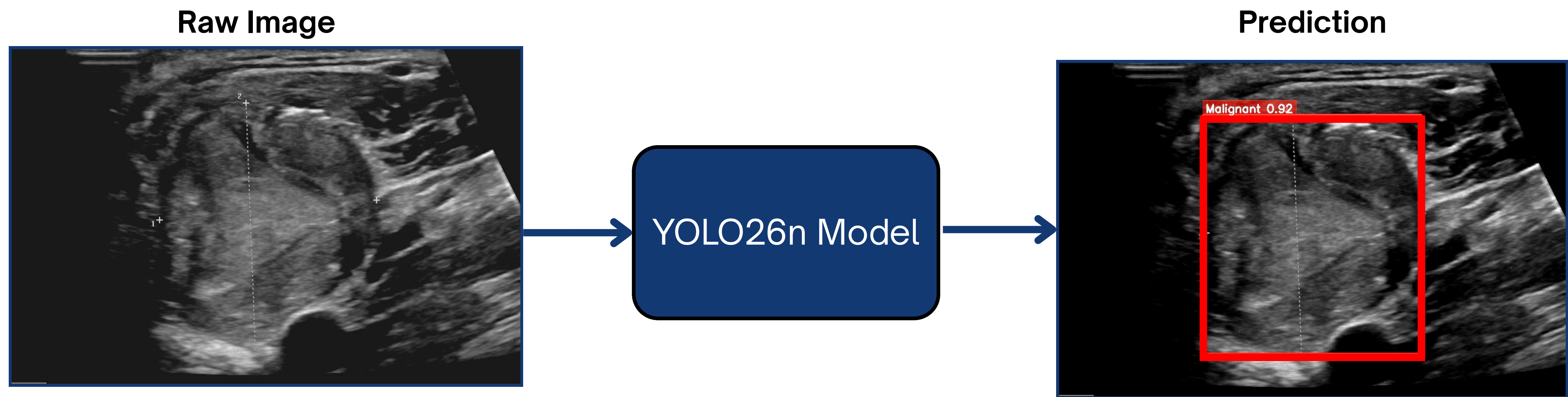
Dataset

- To maximize morphological variance, a dataset comprising **5,266 baseline B-mode ultrasound frames** was curated from three primary repositories.
 - **TN5000 (92.56%):** 4,874 expert-annotated, biopsy-confirmed thyroid scans providing high-fidelity ground truth.
 - **Breast Ultrasound (5.01%):** 264 images to introduce cross-organ morphological features, teaching the network generalized characteristics of malignancy in soft tissue.
 - **TG3K (2.43%):** 128 images to increase exposure to varied pathological presentations.

Training Phase



Inference Phase



Results

Class	Images	Instances	Precision (P)	Recall (R)	mAP@50	mAP@50-95
All (Geral)	519	519	0.87	0.869	0.915	0.63
001131 (Benign)	158	158	0.862	0.793	0.885	0.585
001970 (Malignant)	361	361	0.878	0.945	0.946	0.675